

Credit Card Fraud Detection Analysis

Identifying Fraud Patterns and Designing Real-Time Alert Triggers

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Data Analyst Portfolio Project

1. Business Question

Which transaction patterns consistently signal fraudulent behaviour, and what triggers should be set in a real-time fraud alert system?

This project answers that question using plain, explainable data analysis — no machine learning — so that any finding can be traced back to a specific, provable step in the data.

2. Dataset

The analysis uses the public Kaggle “Credit Card Fraud Detection” dataset: 284,807 anonymized European card transactions, with 31 columns (Time, V1–V28, Amount, Class). After removing 1,081 duplicate rows, 283,726 clean transactions remained, of which 473 (0.17%) were confirmed fraud.

3. Approach

- Cleaned the data — checked for missing values and removed duplicate transactions.
- Measured the basic scale of the problem — fraud rate, and how fraud amounts compare to legitimate ones.
- Ranked all 28 anonymized features by how strongly their average value differs between fraud and legitimate transactions.
- Identified the 3 strongest, non-overlapping signals: V14, V3, and V17.
- Used bucket analysis to find the exact value range (“danger zone”) where fraud concentrates for each.
- Built and tested two rule-based alert triggers — a strict “Tight” rule and a broader “Wide” rule.
- Checked supporting patterns: time-of-day and transaction amount.

4. How Serious Is the Fraud Problem?

Fraud is rare — only 0.17% of transactions — but not random. Fraud amounts show a lower median than legitimate transactions, alongside a higher average, indicating that most fraud attempts are small “test” transactions, with occasional larger ones.

Metric	Value
Total Transactions	283,726
Fraud Transactions	473
Fraud Rate	0.17%
Average Fraud Amount	₹123.87
Median Fraud Amount	₹9.82

[Power BI — Overview page (KPI cards, donut chart, amount comparisons) — paste your Power BI screenshot here]

5. What Patterns Are Associated With Fraud?

5.1 Time of Day

Fraud shows a strong, recurring spike within a specific 2–3 hour window in each 24-hour cycle. Based on how this dataset is typically documented, this window is commonly interpreted as early morning — though the exact clock alignment cannot be independently confirmed from the data alone.

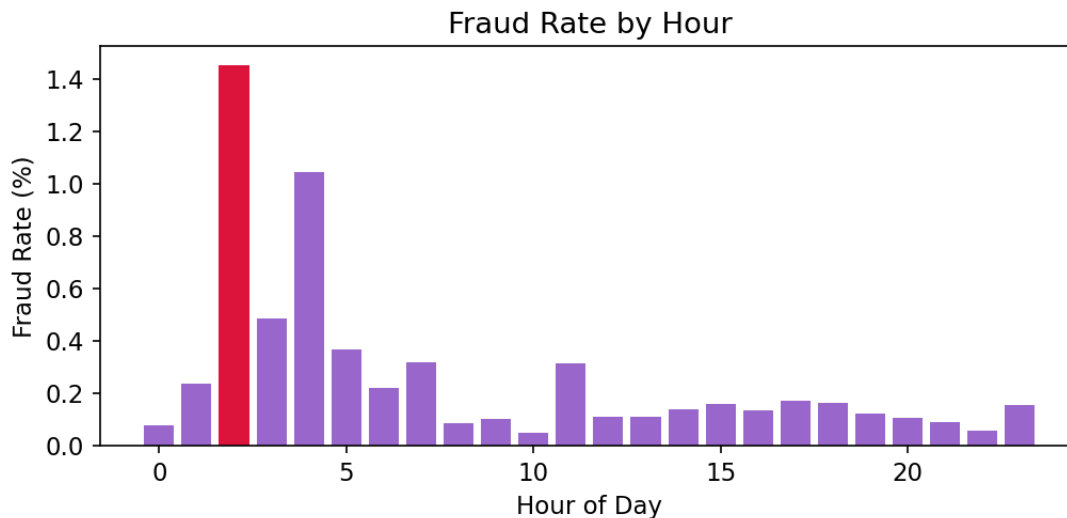


Figure 1: Fraud rate by hour, with the riskiest hour highlighted.

5.2 Feature Danger Zones

Three anonymized features — V14, V3, and V17 — show low correlation with each other (0.06–0.10), meaning each adds genuinely new information. For all three, the most negative value range shows a fraud rate 10–100x higher than every other range.

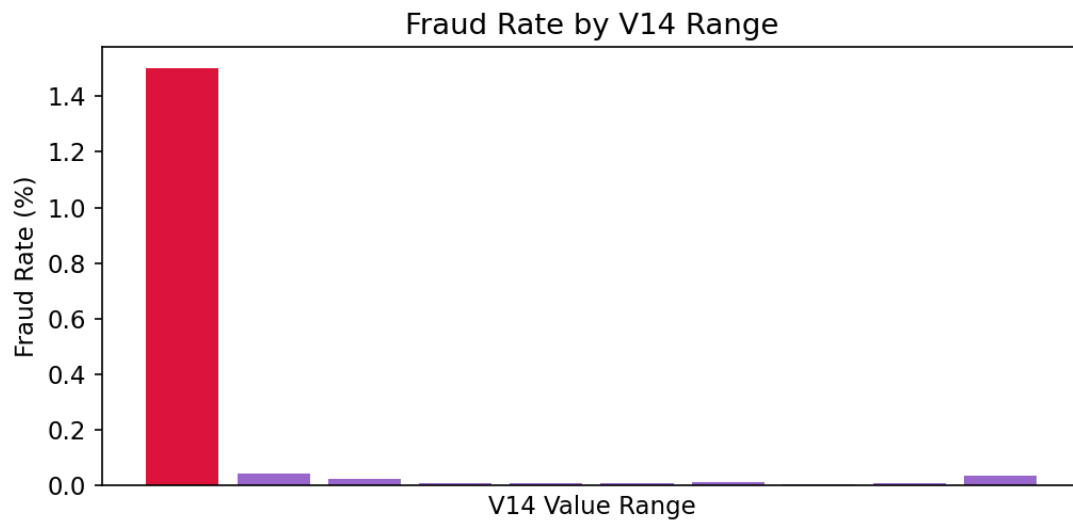


Figure 2: Fraud rate by V14 range — danger zone highlighted.

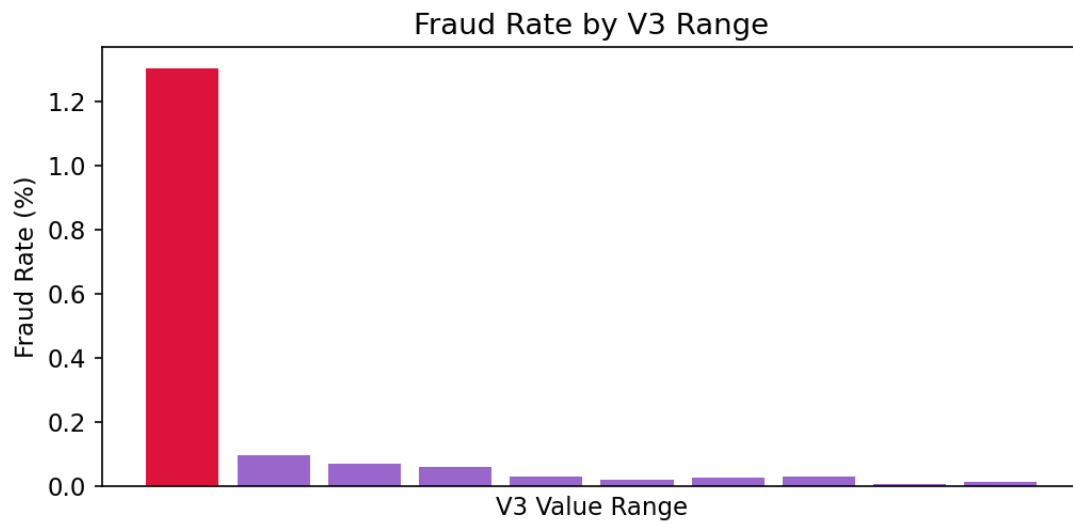


Figure 3: Fraud rate by V3 range — danger zone highlighted.

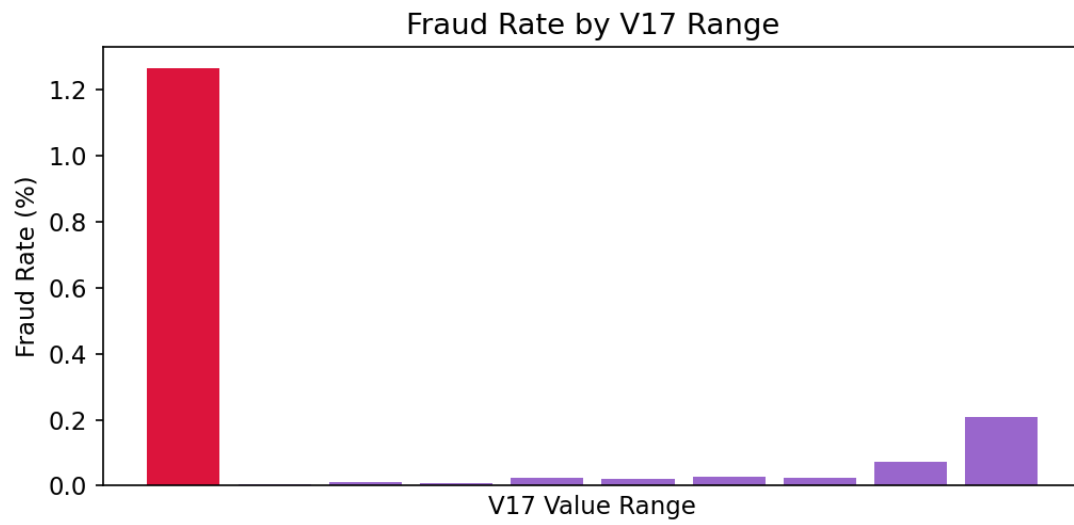


Figure 4: Fraud rate by V17 range — danger zone highlighted.

[Power BI — Fraud Patterns page (all 4 charts together) — paste your Power BI screenshot here]

5.3 Transaction Amount

Fraud transactions have a lower median amount than legitimate ones, but a higher average — most fraud attempts are small, with occasional large outliers pulling the average up.

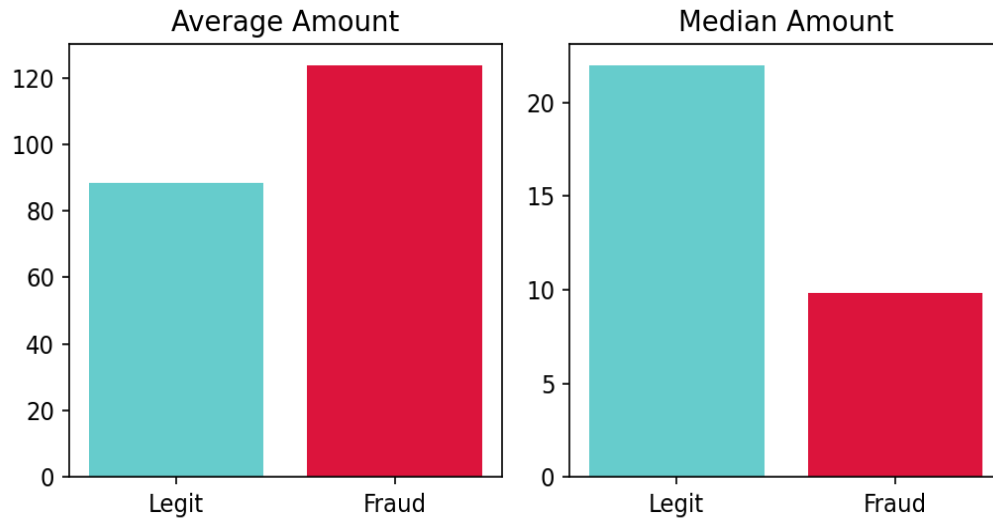


Figure 5: Average vs. median transaction amount, fraud vs. legitimate.

6. What Should a Fraud Team Actually Do?

Two proposed analytical triggers were built and tested against the full dataset, using the danger-zone thresholds identified above.

6.1 Proposed Analytical Triggers

Tight Rule (AND — all three conditions must be true):

$V14 < -1$ AND $V3 < -1.8$ AND $V17 < -0.8$

Wide Rule (OR — any one condition is enough):

$V14 < -1$ OR $V3 < -1.8$ OR $V17 < -0.8$

6.2 Measured Performance

Rule	Total Flagged	Frauds Caught	False Alarms	Catch Rate %	Precision %
Tight (AND)	730	316	414	66.8%	43.29%
Wide (OR)	76,548	439	76,109	92.8%	0.57%

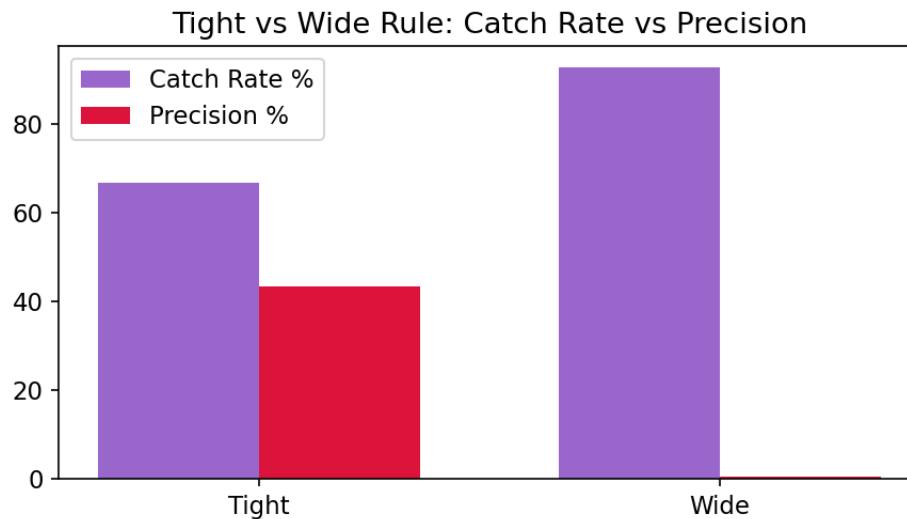


Figure 6: Catch rate vs. precision, Tight vs. Wide rule.

[Power BI — Fraud Alert Rules page (recommendation + comparison table) — paste your Power BI screenshot here]

6.3 Recommendation

No single rule serves both jobs a real fraud system needs. The Tight rule is precise enough (43.29% precision) for automatic transaction holds, with a low false-alarm count. The Wide rule catches far more fraud (92.8%) but is too noisy (76,109 false alarms) for automatic action — it is better suited to a human review queue that flags borderline transactions for closer inspection.

- Use the Tight rule for automatic hold/block — safe, low-noise, high-confidence.
- Use the Wide rule to route borderline transactions to a human analyst — broader safety net.
- Treat the identified high-risk time window as an additional risk booster, especially alongside small transaction amounts.

7. Limitations

- The dataset is from 2013 and one specific region — real-world fraud patterns evolve constantly.
- The Wide rule, as-is, is not realistically deployable due to its high false-alarm volume.
- The real-world meaning of V14, V3, and V17 is unknown by design, since these are privacy-preserving anonymized features (via PCA).
- A production fraud system would combine many more signals (device data, location, transaction history) beyond the three used here.
- These are proposed analytical triggers, not a production-ready system — intended to demonstrate the analytical approach.

8. Why No Machine Learning

Simple, explainable threshold rules are fast to compute, transparent, and easy for a non-technical fraud team to trust and adjust — which matters more for a real-time alert system than squeezing out marginal accuracy gains from a black-box model. Every finding in this report traces back to a specific, reproducible calculation.

9. Conclusion

This analysis identified three explainable, data-driven fraud indicators (V14, V3, V17), quantified their exact danger zones, and translated them into two tested, tradeoff-aware alert rules — directly answering the business question of which patterns signal fraud and what triggers a real-time system should use.